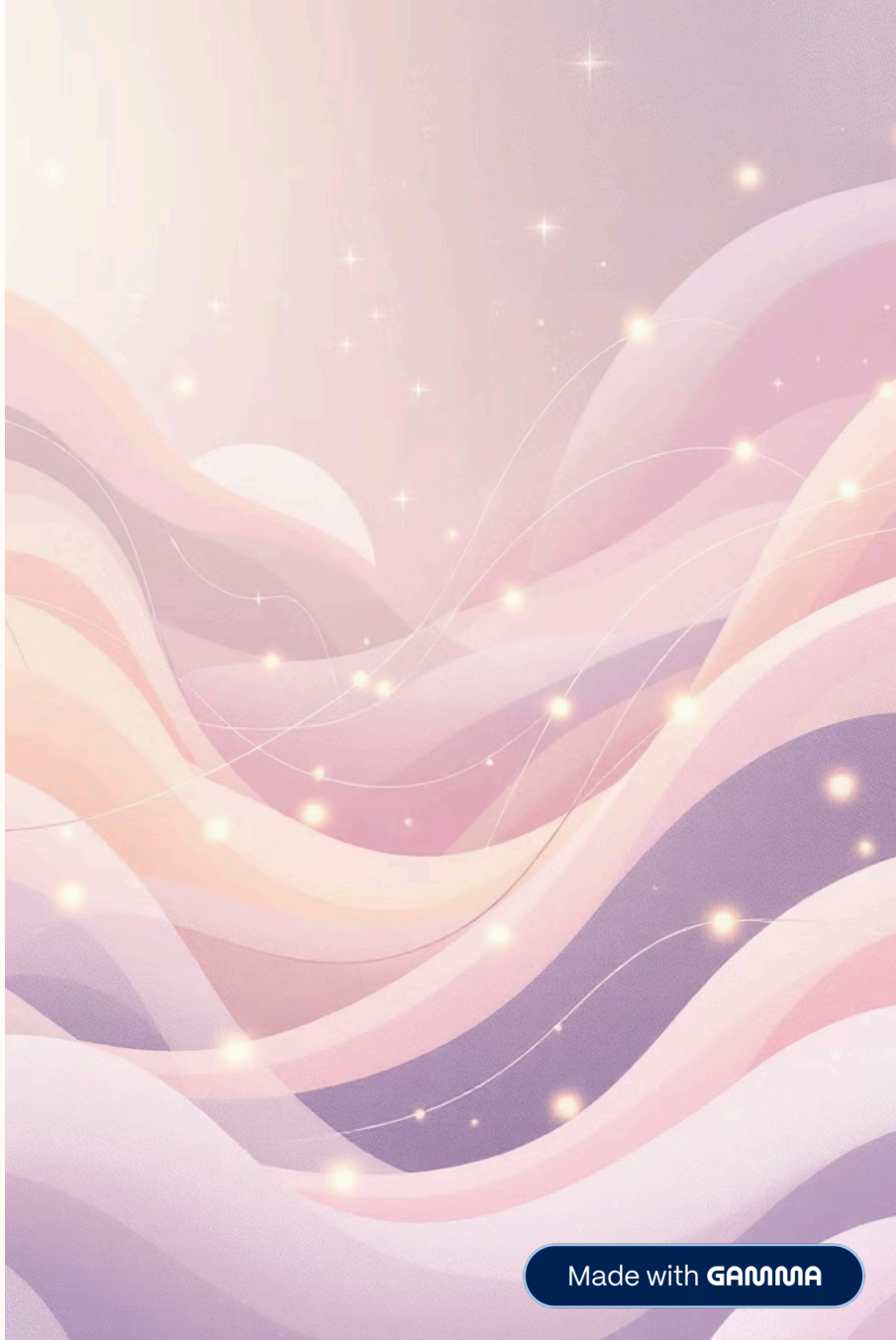
Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.



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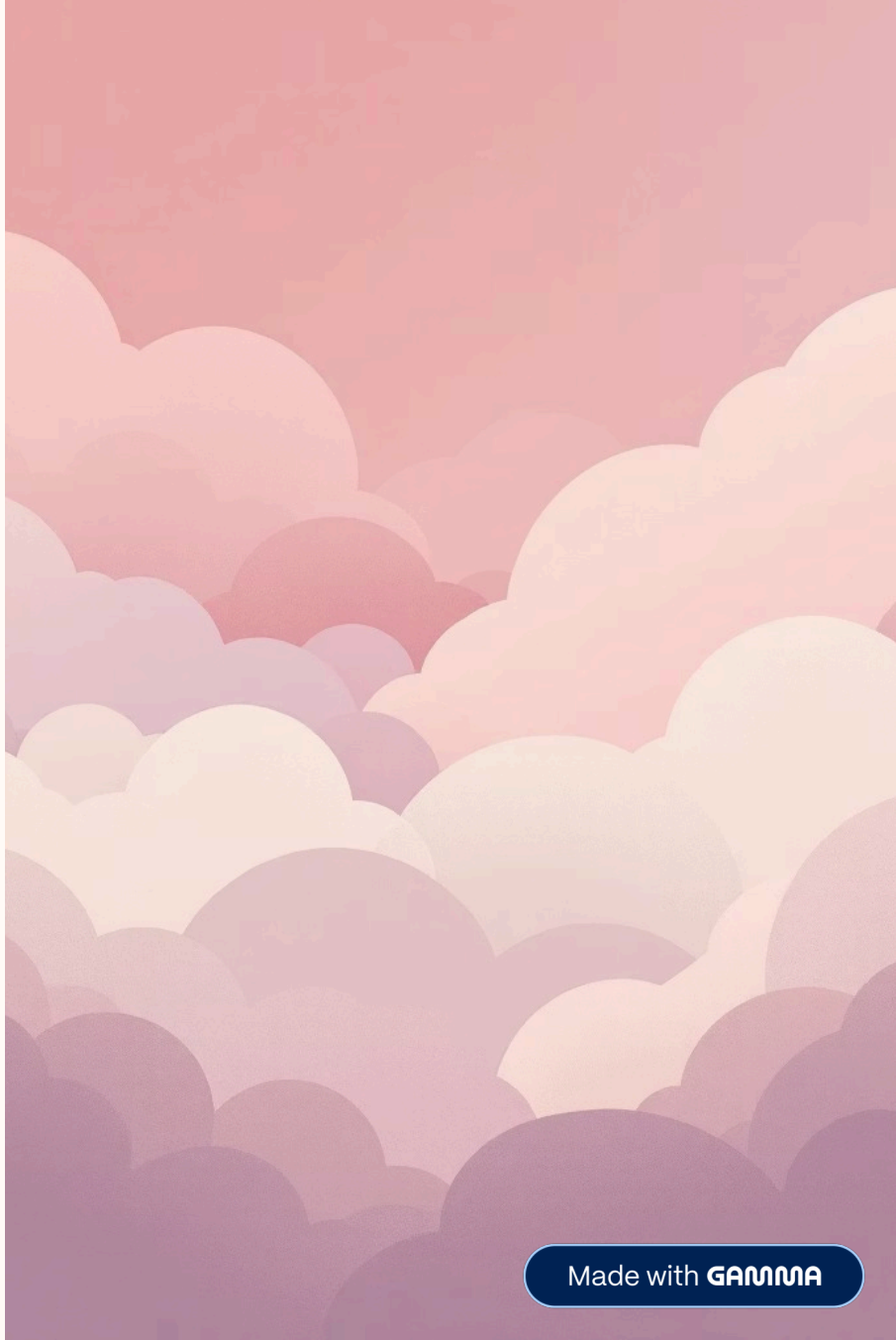
Project Overview & Dataset

Project Goal

Analyze 3,900 purchases to optimize operations, understand preferences, and segment customers.

Dataset Snapshot

- 3,900 Rows, 18 Columns
- Demographics, Purchase Details, Behavior
- Missing: 37 Review Ratings



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Exploratory Data Analysis (Python)

Data Loading & Exploration

Imported with pandas, checked structure and summary statistics.

Data Cleaning

Imputed missing 'Review Rating' with median per category.

Feature Engineering

Created 'age_group' and 'purchase_frequency_days'.

Database Integration

Cleaned data loaded into PostgreSQL for SQL analysis.

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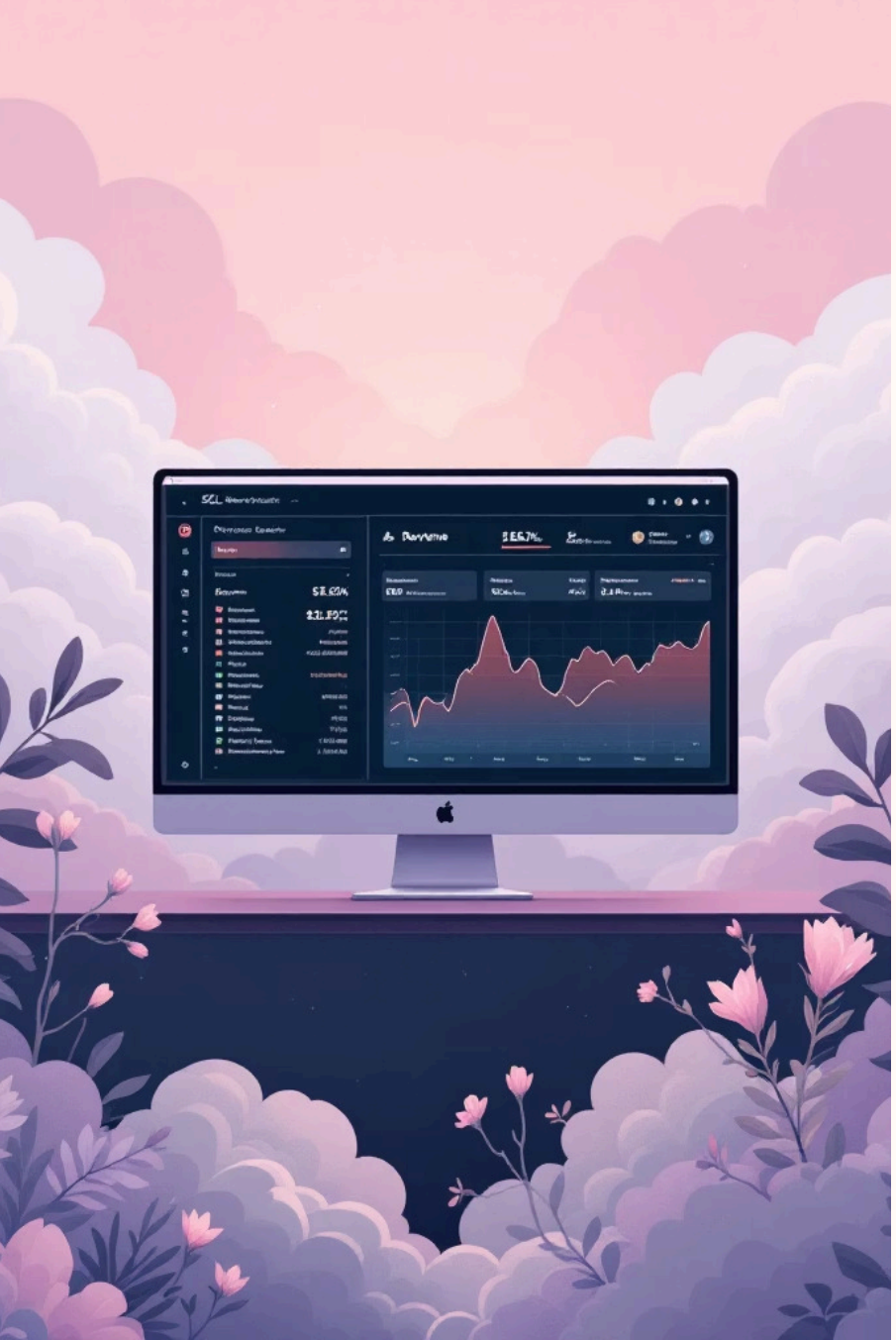

Key SQL Analysis Findings

Revenue by Gender

Male customers generated significantly more revenue ($157,890 vs $75,191).

Top 5 Products by Rating

Gloves, Sandals, Boots, Hat, Skirt had highest average review ratings.

Shipping Type Comparison

Express shipping had higher average purchase amounts ($60.48 vs $58.46).

Subscribers vs. Non-Subscribers

Non-subscribers contributed more total revenue ($170,436 vs $62,645).

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Customer Segmentation & Product Insights

Customer Segments

- Loyal: 3116
- Returning: 701
- New: 83

Discount-Dependent Products

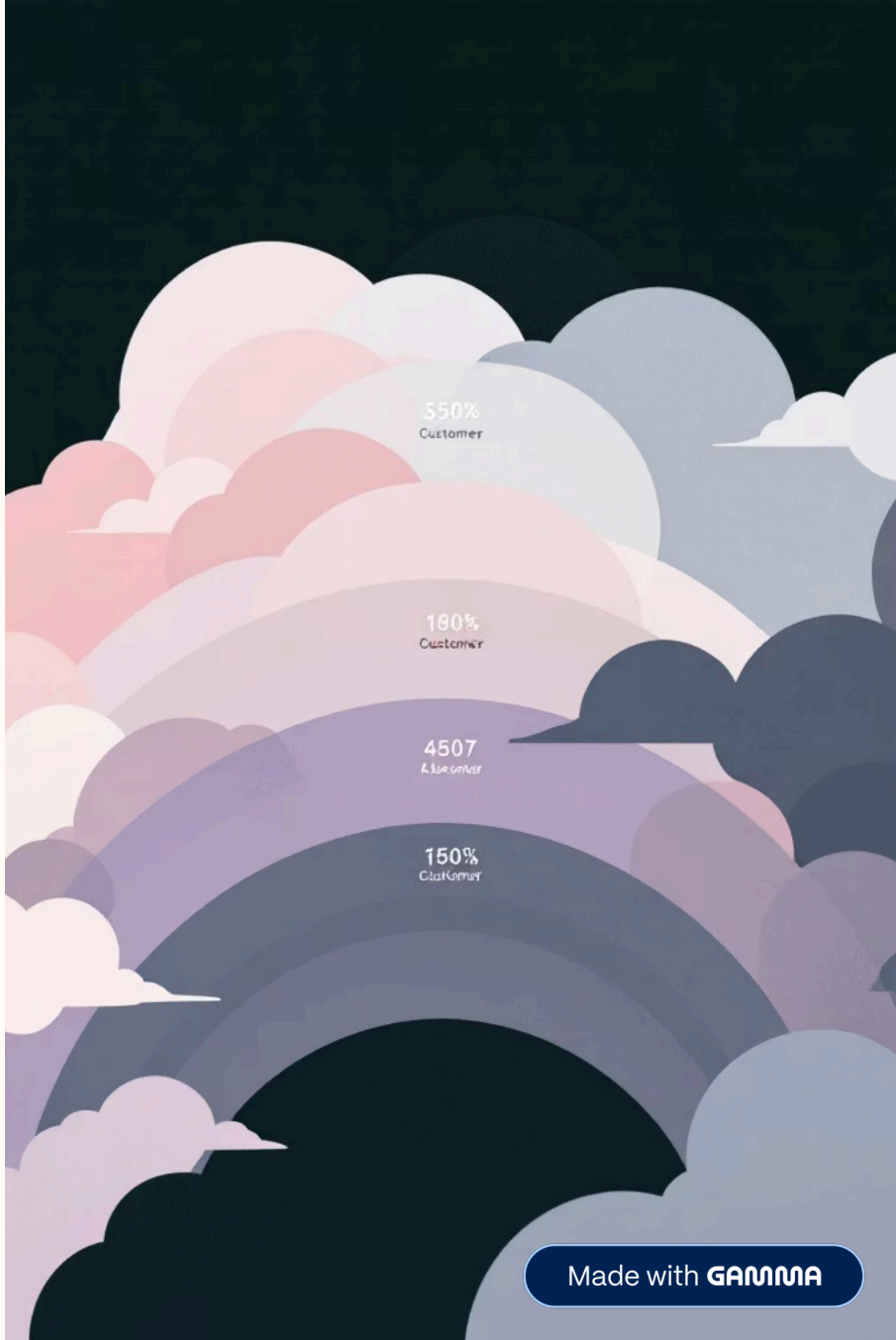
Hat, Sneakers, Coat, Sweater, Pants had highest discount rates.

Top Products per Category

Jewelry (Accessories), Blouse (Clothing), Sandals (Footwear), Jacket (Outerwear) were top sellers.

Repeat Buyers & Subscriptions

2518 non-subscribers vs 958 subscribers among repeat buyers.



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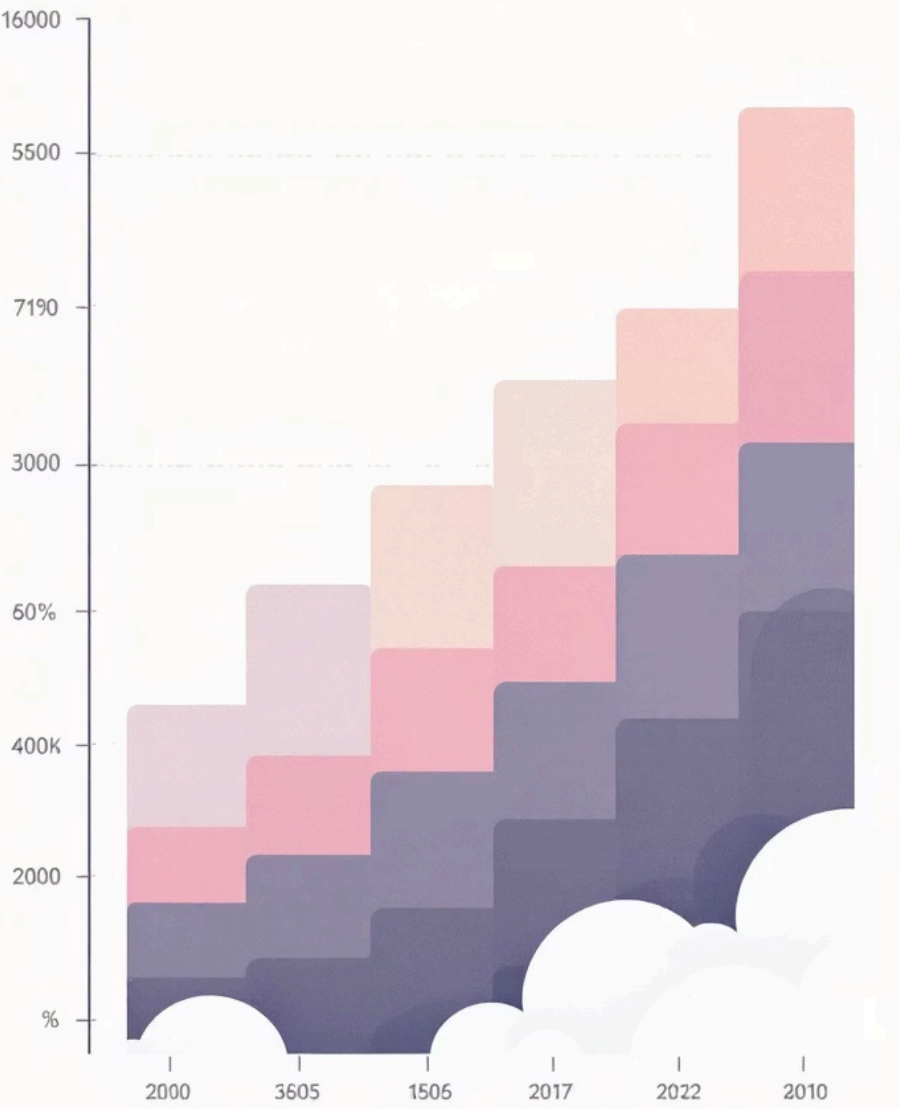
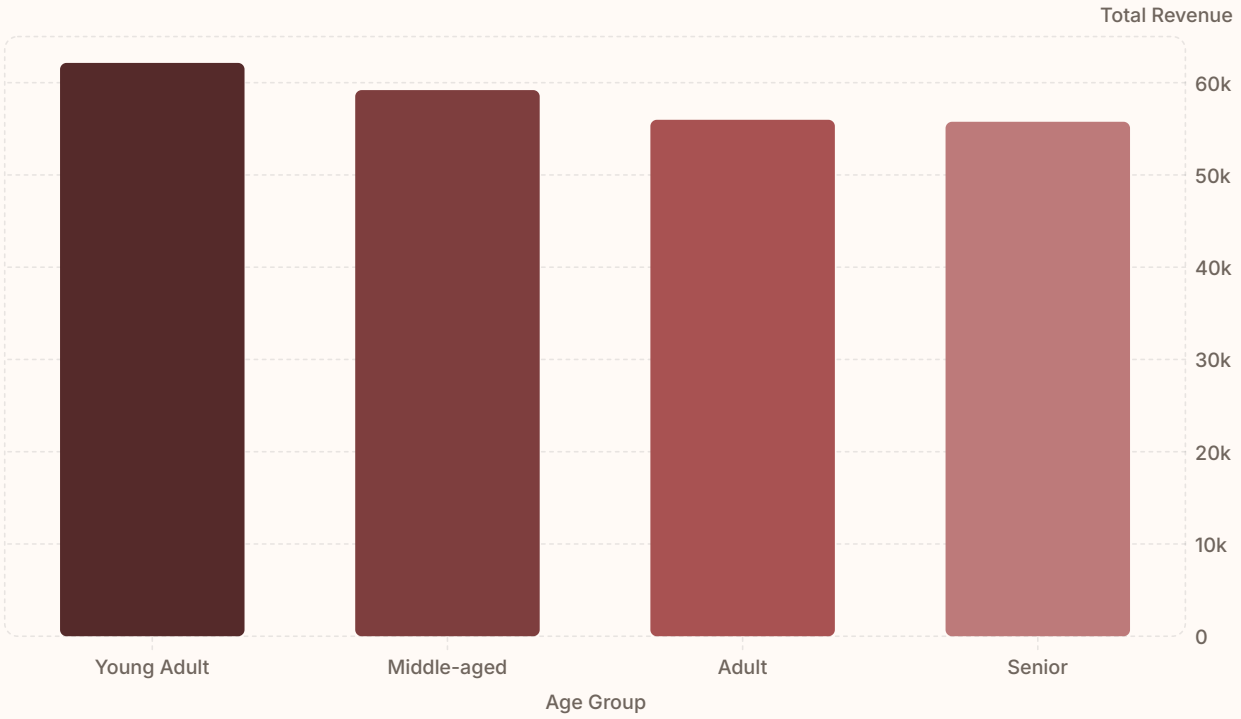


Revenue by Age Group



Young Adults generated the highest revenue, followed by Middle-aged customers.

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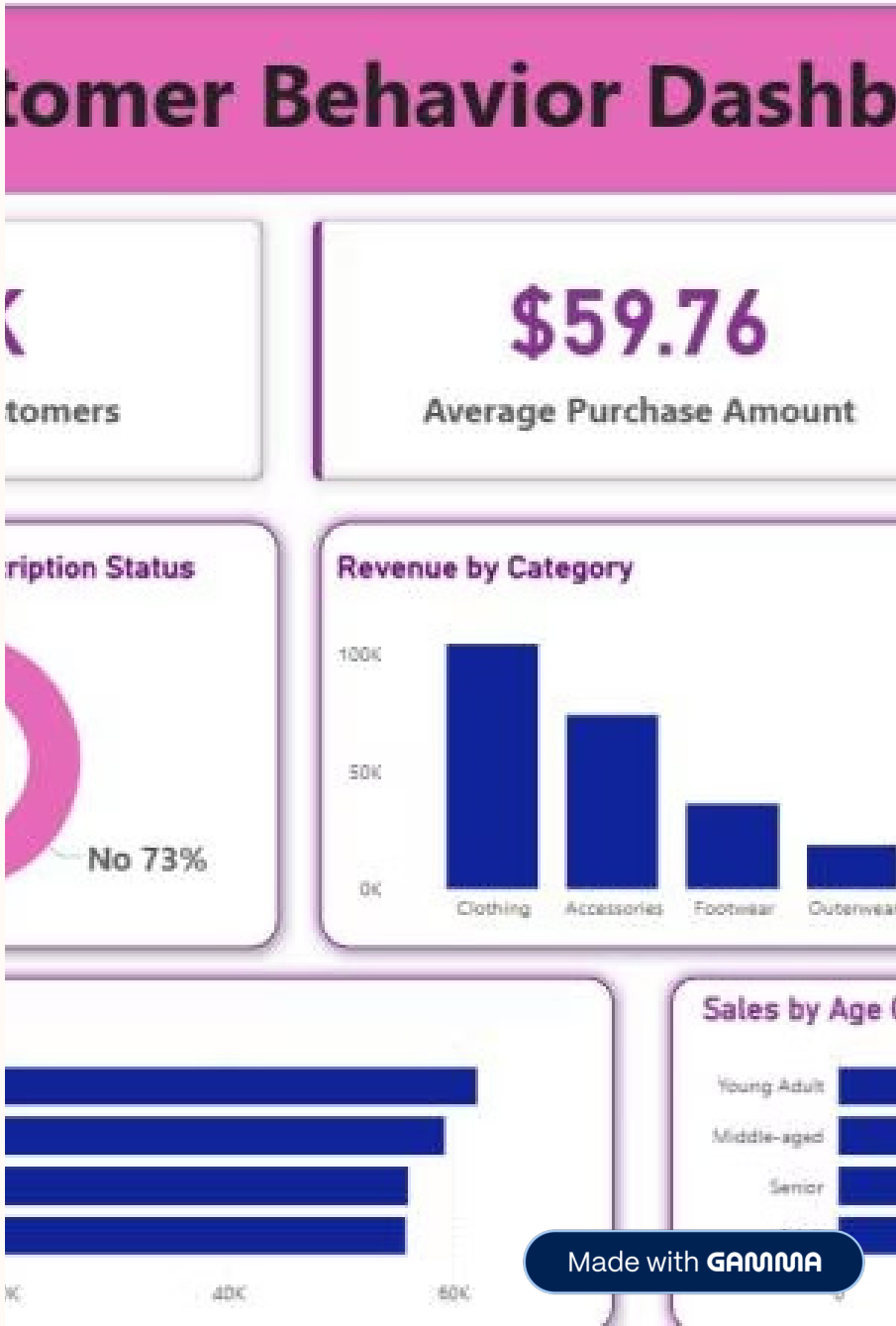
Interactive Power BI Dashboard

Visualizing key metrics and insights for dynamic exploration.

3.9K	$59.76	3.75
Customers	Avg. Purchase	Avg. Review



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Key Business Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber base.

Customer Loyalty Programs

Reward repeat buyers to foster loyalty and move to "Loyal" segment.

Review Discount Policy

Optimize discounts for sales boost without sacrificing margins.

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.

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